



JKBOSE 2026

CLASS 11TH

PERMUTATION AND COMBINATIONS

MATHS

LECTURE-07

BY-MEHRAJ SIR



TOPICS TO BE COVERED

AB - - - AD - -

1 Lexicographic order Problems (Dictionary order problems)

2 Interview Problems

3 Gap Method Problems

A, B, C

ABC, ACB, BCA, BAC, CBA, CAB

↓
ABC
ACB
BAC
BCA
CAB
CBA

EXAMPLE

A | • | • | • | •

RAD $\frac{I}{P}$ $\frac{P}{I}$



If all the letters of the word 'RAPID' are arranged in all possible manner as they are in a dictionary, then find the rank of the word 'RAPID'?

(Sol) Alphabetical order of the letters of word "RAPID" is : A, D, I, P, R

Here, no. of words starting with A = $4! = 24$

No. of words starting with D = $4! = 24$

No. of words starting with I = $4! = 24$

No. of words starting with P = $4! = 24$

Now, No. of words starting with the RAD = $2! = 2$

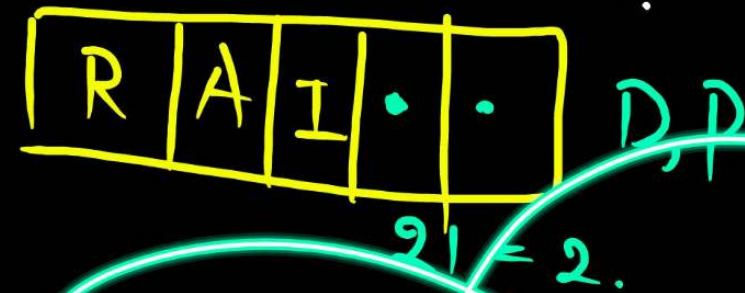
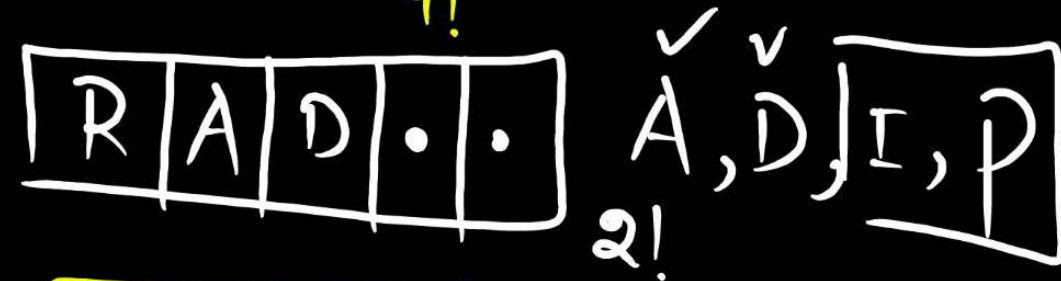
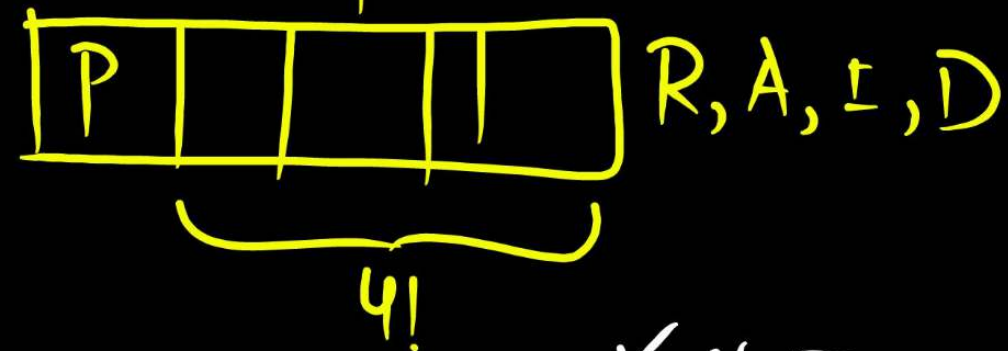
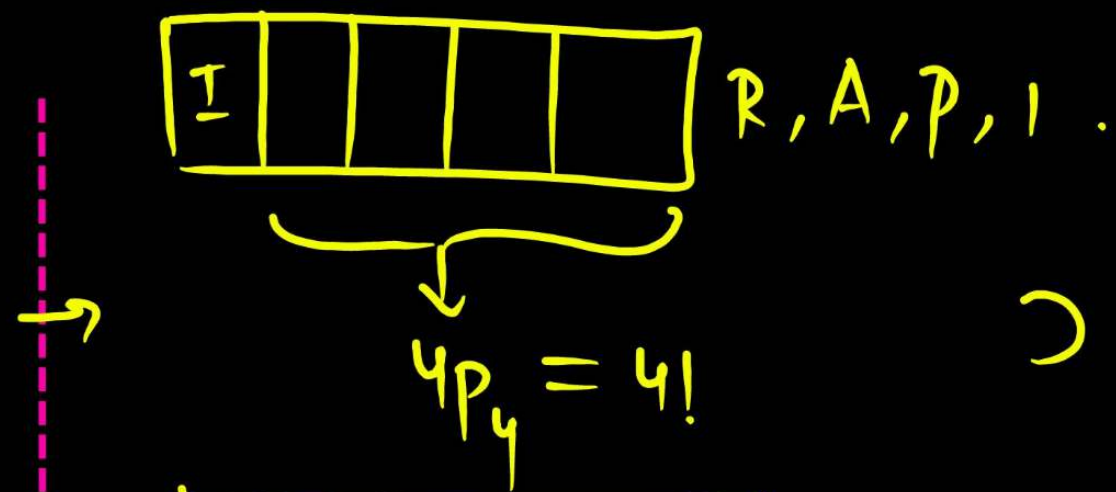
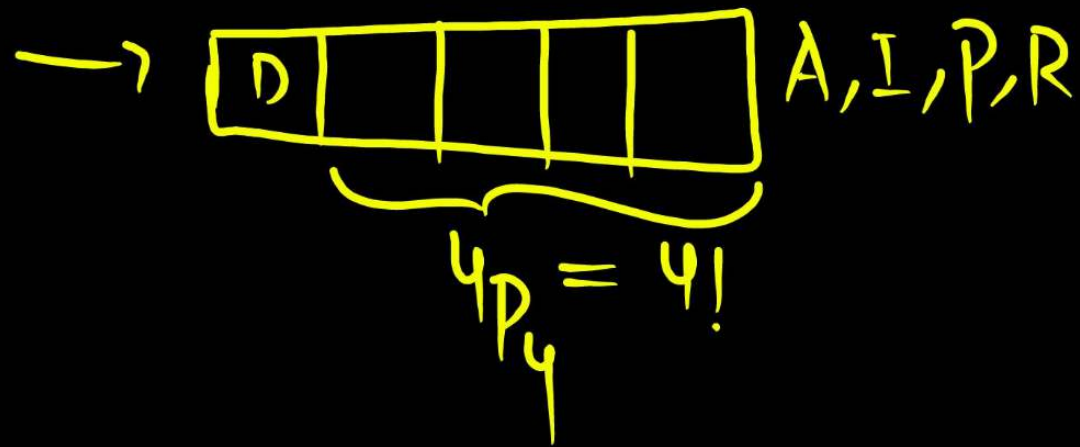
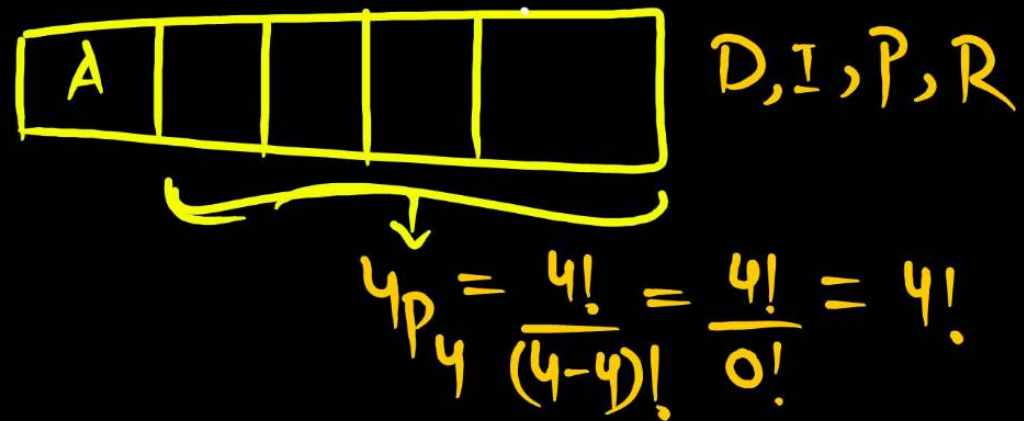
No. of words starting with the RA I = $2! = 2$

Therefore the word at 10th rank is RAPDI.

So, the rank of the word RAPID = 102

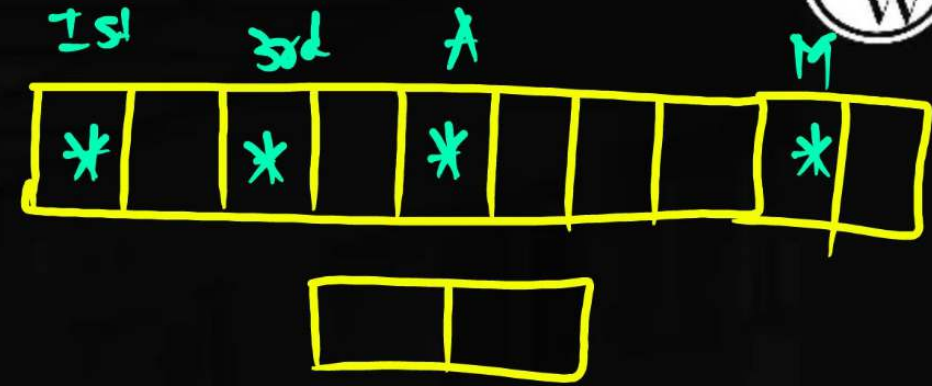
A | | | |
RDIP

Hence the result.



EXAMPLE

$${}^8P_8 = \frac{8!}{(8-8)!} = \frac{8!}{0!} = \frac{8!}{1} = 8! = ({}^8P_8)$$



In how many ways interview of 10 people can be made, if

- (i) Aksha De comes before Mutiaba.
- (ii) Aksha De comes before Mutiaba and Mutaiba comes before Para Special force
- (iii) Aksha De comes before Mutiaba and Para Special force.

(Sol) (i) Required No. of ways
 $= ({}^{10}C_2 \times 1) \times 8! = \underline{\hspace{2cm}}$

(ii) Required No. of ways
 $= ({}^{10}C_3 \times 1) \times 7! = \underline{\hspace{2cm}}$

(iii) Required No. of ways
 $= ({}^{10}C_3 \times 2!) \times 7! = \underline{\hspace{2cm}}$

A - M - P ✓
A - P - M ✓

EXAMPLE

A bag contains 50 tickets numbered 1, 2, 3, 4, ... 50. Find the number of set of five tickets x_1, x_2, x_3, x_4, x_5 one has if $x_1 < x_2 < x_3 < x_4 < x_5$ and $x_3 = 30$.

(Homework)

$$\binom{29}{2} \times 1 \quad \text{or} \quad \binom{29}{2} \times 1$$

#

QUESTION(JEE-MAINS 2026)

The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is

1

1579 ✗

2

1581 ✗

3

1578 ✗

4

1580 ✓

(Sol) Alphabetical order of the given letters is: A, D, P, R, U, U, Y

$$\begin{aligned} \text{Required number} &= \frac{6!}{2!} + \frac{6!}{2!} + \frac{6!}{2!} + \frac{6!}{2!} + 5! \\ &\quad + 3! + 3! + 3! + 1 + 1 \\ &= (360 \times 4) + 120 + 18 + 1 + 1 \\ &= 1580. \end{aligned}$$

UA _ _ _ _
D, P, R, U, Y

UDAP _ _ _
U, R, Y

UDAR _ _ _
U P Y

UDAU _ _ _
P, R, Y

UDAYR U, P, R
UDAYPRU
UDAYPUR

✓ ✓ ✓ ✓
A, D, P, R, u, u, y

$$\left(4 \times \frac{6!}{2!}\right) + 5! + 3! + 3! + 3!$$

UDAP _ _ _

UDAR _ _ _

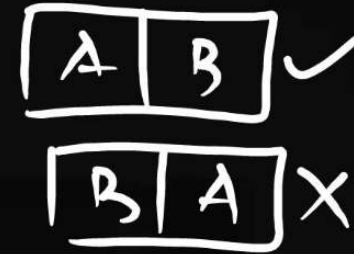
UDAU _ _ _

UDAYPRu 1579

UDAYPUR → 1580.

QUESTION

In how many ways can **5 people** (A, B, C, D, E) be arranged for an interview if **A** must always **come before B**?



1

120 ✗

2

60 ✓

3

240 ✗

4

40 ✗

(Sol) Required No. of ways
 $= ({}^5C_2 \times 1) \times 3!$

$$= \frac{5!}{2!3!} \times 3! = \frac{5 \times 4 \times 3 \times 2}{2!} = 60$$

QUESTION

If **6 candidates** are being **interviewed**, in how many ways can the schedule be made such that Candidate **X** comes before **Y**, and Candidate **Y** comes before **Z**?

1 ✓ 120

2 ✗ 20

3 ✗ 360

4 ✗ 720

(Sol) Required No. of ways

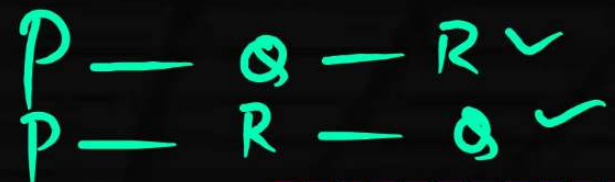
$$= ({}^6C_3 \times 1) \times 3!$$

$$= \frac{6!}{3!3!} \times 3! = \frac{6 \times 5 \times 4 \times 3!}{3!}$$

$$= 20 \times 6$$

$$= 120$$

QUESTION



In a group of 10 people, how many arrangements exist where Person P is interviewed before both Person Q and Person R?

1 $\frac{10!}{3!} \times ({}^{10}C_3 \times 2!) \times 7!$

2 $\frac{10!}{2} = \frac{10! \times 2! \times 7!}{7! \times 3!}$

3 $\frac{10!}{6!} \times \frac{10! \times 2!}{3 \times 2!}$

4 $\frac{10!}{3} = \frac{10!}{3}$

QUESTION

In a race of **8 runners**, how many finishing orders are possible where **Runner A** finishes in the top **3** and Runner **B** finishes somewhere after Runner **A**?

- 1** 12960 (homework)
#
- 2** 6480
- 3** 12900
- 4** 6460

QUESTION

In an interview of **10 people**, how many ways can it be done if **Person A is before B**, and **Person C is before D**, and **Person E is before F**?

- 1** $\frac{10!}{(2!)^3}$ ✓
- 2** ✗ $\frac{10!}{(2!)^3} \times 3! \times 4!$
- 3** ✗ $\frac{10!}{(2!)^3} \times 3!$
- 4** ✗ $\frac{10!}{2!} \times 3!$

(Sol) Required No. of Selections

$$= ({}^{10}C_2 \times 1) \times ({}^8C_2 \times 1) \times ({}^6C_2 \times 1) \times 4!$$

$$= \frac{10!}{2!8!} \times \frac{8!}{2!6!} \times \frac{6!}{2!4!} \times 4!$$

$$= \frac{10!}{(2!)^3}$$

QUESTION (JEE MAINS 2023)

The letters of the word "OUGHT" are written in all possible ways and these words are arranged as in a dictionary, in a series. Then the serial number of the word "TOUGH".

☒ 1 89

☐ 2 84

☐ 3 86

☐ 4 79

Alphabetical order:

G, H, O, T, U

$$\begin{aligned} \text{Required No.} &= (4! \times 3) + 3! + 3! + 2! + 2! + 1 \\ &= (24 \times 3) + 17 \\ &= 72 + 17 \\ &= 89 \end{aligned}$$

TG _ _ _

TH _ _ _

TOG _ _

TOH _ _

TOUGH H

QUESTION

What is the rank of the word '**MOTHER**' when all its letters are arranged in alphabetical order to form dictionary words?

1 420

Homework
#

2 309

3 509

4 307

QUESTION (Miscellaneous Example)

In how many ways can 5 girls and 3 boys be seated in a row so that no two boys are together?

Home work.

$$* G_1 * G_2 * G_3 * G_4 * G_5 *$$

$$5! \times {}^6C_3 \times 3!$$

$$= 5! \times {}^6P_3$$

QUESTION

In how many ways can **6 boys** and **4 girls** be arranged in a row such that **no two** girls are **together**?

1

$$6! \times 7P_4$$



2

$$10! - 4! \times$$

3

$$6! \times 4! \times$$

4

$$7! \times 6P_4 \times$$

$$6! \times 7C_4 \times 4! \\ = 6! \times 7P_4$$

$$*B_1 * B_2 * B_3 * B_4 * B_5 * B_6 *$$

QUESTION

There are **5 red balls** and **3 blue balls** (all identical within their color). In how many ways can they be arranged in a row so that **no two blue balls** are together?

1 20 ✓

* R * R * R * R * R *

2 120 ✗

$$\frac{5!}{5!} \times {}^6C_3 \times \frac{3!}{3!}$$

3 10 ✗

$$= 1 \times {}^6C_3 \times 1$$

4 60 ✗

$$= \frac{6!}{3!3!} = \frac{6 \times 5 \times 4 \times 3!}{\cancel{3!} \times \cancel{3!}} = 20.$$

QUESTION

If **7 people** are to be seated in a row, including A and B, in how many ways can they be seated such that **A and B** are never adjacent?

1 1440

Homework.

2 3600

3 5040

4 2520

QUESTION

In a row of **10 chairs**, **3 people** are to be seated such that **no two** people sit in consecutive chairs. How many such arrangements are possible?

1 720

2 120

3 336

4 210

Homework.

QUESTION

How many words can be formed using all letters of the word '**EQUATION**' so that no **two vowels** are together?

1 0

2 40320

3 2880

4 1440

Homework.

QUESTION

8 people are in a row. How many ways can they be arranged so that **3 specific people** (X, Y, Z) are **not all three** together, but **any two** of them can be together?

1 $8! - 6!$

2 $\frac{8!}{3!}$

Homework.

3 $8! - (6! \times 3!)$

4 $5! \times {}^6P_3$

Thank You

GRID Problem

Candy Flavour Problem